



## EMERGENCE SIGNAL CHECK: EIGHT BRAIN-HEALTH TOPICS

Stage 1–3 internal note — Google Trends and OpenAlex only, single collection round, 21–22 July 2026

### THE BRAIN HEALTH PRIORITY GAP PROJECT — TECHNICAL BRIEFING

### STAGE 1–3 RESULTS

This is not the briefing note. It reports emergence scores from the first Stage 1–3 collection pass only. It does not compute or claim a Brain Health Priority Gap Index — that requires MeSH alignment (Stage 4), which has not been run. Read this as a decision input for whether to proceed to Stage 4, not as a finding about recognition gaps.

**All eight candidate topics show positive emergence on a single collection round; four show markedly higher emergence than the rest.** Perimenopause and memory, long COVID cognitive effects, ADHD in women and neurodiversity (adult-diagnosed) combine rising Google Trends search interest with rising OpenAlex publication counts at the same time — the convergence the emergence-score formula is designed to require, per methodology.md Section 5. The remaining four topics are still growing on both signals, just less sharply.

Emergence score is calculated exactly as pre-registered: clipped Google Trends growth (recent 3-month mean vs. prior 12-month mean) multiplied by clipped OpenAlex publication growth (trailing 3-year works vs. prior 3-year works from a 6-year series), both terms floored at zero before multiplying<sup>1</sup>. No window or formula choice was adjusted after seeing these numbers.

**Executive function support is a known outlier, not a finding.** Its OpenAlex counts (80,067 prior-period works, 97,559 recent) are an order of magnitude larger than every other topic's, because both of its keyword variants ("executive function", "executive function support") are generic clinical terms without a women's-health qualifier — flagged in methodology.md Section 6 as a candidate-definition limitation before this data was collected, not adjusted after the fact.

8 / 8

TOPICS WITH POSITIVE EMERGENCE

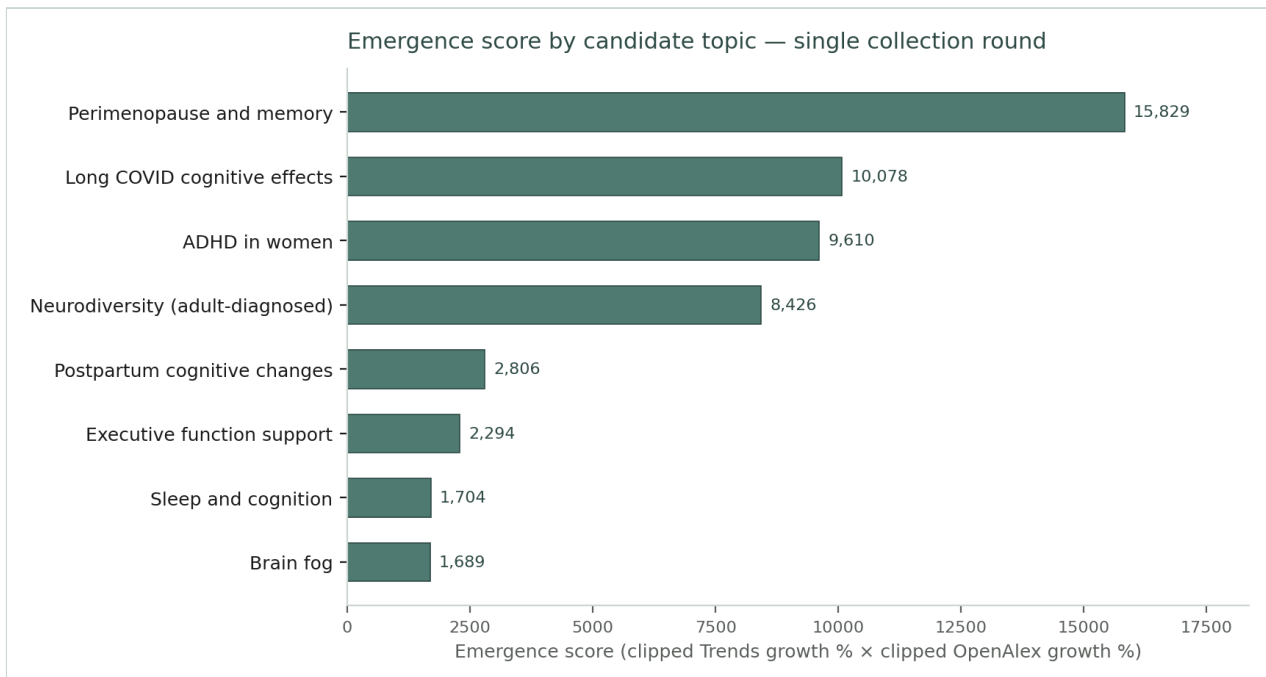
4

TOPICS SCORING ABOVE 8,000

#### WHAT THIS NOTE DOES NOT SHOW

- No MeSH alignment — Stage 4 not run.
- No Priority Gap Index — requires Stage 4.
- No trend line — one collection round only.

FIGURE 1 EMERGENCE SCORE BY CANDIDATE TOPIC



Source: own analysis, Google Trends (pytrends) / OpenAlex Works API, 21–22 July 2026.

(1) Both growth terms are clipped at zero before multiplying so that two co-declining signals cannot produce a false-positive emergence score — documented in calculate\_topic\_signals.py and methodology.md Section 5 before this data was collected.

# EVIDENCE TABLE AND DECISION ASK

Stage 1–3 internal note — continued

## EVIDENCE SUMMARY

EMERGENCE COMPONENTS BY TOPIC, RANKED BY EMERGENCE SCORE

| Topic                            | Trends growth % | OpenAlex growth % | Emergence score |
|----------------------------------|-----------------|-------------------|-----------------|
| Perimenopause and memory         | 117.8           | 134.4             | 15,829          |
| Long COVID cognitive effects     | 121.7           | 82.8              | 10,078          |
| ADHD in women                    | 100.1           | 96.0              | 9,610           |
| Neurodiversity (adult-diagnosed) | 67.6            | 124.6             | 8,426           |
| Postpartum cognitive changes     | 133.3           | 21.1              | 2,806           |
| Executive function support       | 105.0           | 21.8              | 2,294           |
| Sleep and cognition              | 217.9           | 7.8               | 1,704           |
| Brain fog                        | 53.1            | 31.8              | 1,689           |

## READING THE RANKING CORRECTLY

Sleep and cognition has the single highest Trends growth figure (217.9%) but ranks near the bottom on emergence, because its OpenAlex growth is comparatively flat (7.8%) — the multiplicative formula is doing exactly what it was designed to do: reward convergence across both signals, not strength on either one alone. The same logic explains why postpartum cognitive changes (highest Trends growth of all eight, at 133.3%) still ranks fifth.

This ranking is a prioritisation signal for a second collection round, not a settled result. A single snapshot cannot distinguish a genuine emerging pattern from short-term attention (e.g. a news cycle affecting search interest, or a conference cycle affecting publication counts in a given window).

### DECISION ASK

Per methodology.md Section 8: proceed to Stage 4 (MeSH alignment) for the four highest-emergence topics, or hold all eight for a second collection round first. Both are defensible; this note does not recommend one over the other.